

Sinclair ZX Spectrum Emulator User Guide

ZX SPECTRUM 16K/48K: STARTUP OPTIONS AND
COMMAND CONSOLE REFERENCE

IGNACIO CEA FORNIÉS

Contents

CONTENTS 1

/O 6

/P 6

/H 6

/F 7

/K 7

/C 7

/L 8

/A 8

/D 8

/S 9

/I 9

/R 9

/G 10

/N 10

/W 10

/B 11

/M 11

/H 12

/F 12

/K 12

/C 13

/L	13
/A	13
/D	14
/S	14
/I	14
/R	15
/P	15
/G	15
/N	16
/W	16
/B	16
/M	17
3.1 STANDARD FRAMEWORK COMMANDS	18
HELP	18
AUTHOR	18
STATUS	18
PC	19
REGISTERS	19
CHANGereg	19
STACK	20
CPUSTATUS	20
CPUSSTATUS	20
CPUSTATE	21
CPUINFO	21
MEMORY	21
MEMORYDUMP	22
MEMORYSTR	22
SETMEMORY	22
MEMORYANALYSIS	23
STOP	23
RUN	24
SETPC	24
NEXT	24
NEXTSTACK	25
SHOWNEXT	25
INST	25
BREAKS	26
SETBREAK	26

REMOVEBREAK	26
REMOVEBREAKS	27
SPEED	27
LOADBIN	27
SAVEBIN	28
DEBUGON	28
DEBUGOFF	28
IDEBUGON	29
IDEBUGOFF	29
RESTART	29
DEVICES	30
PERIPHERALS	30
PERCOMMAND	30
ASSIGNJOY	31
CLOCKFACTOR	31
SOUNDON	31
SOUNDOFF	32
INTERRUPTS	32
INTSET	32
CHIPS	33
CRTON	33
CRTOFF	33
DATASETTE	34
GRIDON	34
GRIDOFF	34
PICTURE	35
SETHOOK	35
REMOVEHOOK	35
HOOKS	36
HOOKSHELP	36
CMOVEPARAMS	36
 <u>3.2 INHERITED SINCLAIR COMMANDS</u>	 37
 SYSVARS	 37
SYSVAR	37
 <u>3.3 ZX SPECTRUM-SPECIFIC COMMANDS</u>	 38
 ULA	 38
ULAEVENTS	38
SCREENDUMP	38
COLORDUMP	39
CHARSDRAW	39
 <u>3.4 LOCAL-CONSOLE-ONLY COMMANDS</u>	 39
 LOADPRG	 39
LOADBINARY	40
LOADBLOCKS	40
DECOMPILE	40
POSSIBLEPERS	41
CONNECTPER	41
DISCONNECTPERS	42

LOADPERDATA	42
EMPTYPERDATA.....	42
SAVEPERDATA	43
CLEARPERDATA.....	43
 APPENDIX A. COMMAND AVAILABILITY SUMMARY	 44
 APPENDIX B. SOURCE-OF-TRUTH MAINTENANCE RULE	 44
 ILLUSTRATIONS	 45
 EQUATIONS.....	 46

IGNACIO CEA FORNIÉS

Foreword

This guide describes the startup and command interfaces implemented by the two Sinclair ZX Spectrum executables in EMULATORS: ZX SpectrumEmulator (without a local command console) and ZX SpectrumEmulatorC (with a local command console). It covers the standard 16K/48K machine, PAL and NTSC video, four ROM languages, and is based on both entry points, the ZX SPECTRUM::ZX SpectrumEmulator inheritance chain, LocalConsole, and the complete ZX SPECTRUM -> SINCLAIR -> StandardCommandBuilder responsibility chain.

Command and parameter names are shown exactly as accepted by the current code. Startup options use /XVALUE with no space between the one-letter option and its value. Command names are matched as written by the builders; use uppercase names for reliable operation. Positional command parameters are separated by spaces; examples use paths without embedded spaces.

The non-console executable does not expose an interactive prompt. When started with /o it creates a remote communication system using the same ZX Spectrum, Sinclair and standard command builders. Commands marked Local console only are intercepted by LocalConsole and therefore cannot be sent to the non-console executable.

1. Startup Parameters: Version Without a Local Console

Executable: ZXSpectrumEmulator.exe. It accepts the framework and ZX Spectrum-specific options below, plus /o and the communication-port option. Option letters and language values are case-sensitive because the command-line parser preserves them exactly.

/o

Meaning: Controls creation of the remote command communication system.

General form:

```
/o[YES|NO]
```

- YES or an empty value: open remote communications.
- NO (and any other non-YES non-empty value): do not open it.
- Omitted: remote communications disabled.

Examples:

```
ZXSpectrumEmulator.exe /o  
ZXSpectrumEmulator.exe /oYES /p60000
```

Consequence: When enabled, remote clients can execute ZX Spectrum, Sinclair and standard builder commands. LocalConsole-only commands remain unavailable.

/p

Meaning: Selects the TCP listening port used by the non-console executable's remote command channel.

General form:

```
/pPORTNUMBER
```

- 0 through 65535: requested listening port; values above 1000 are recommended.
- Omitted: port 60000 is used when /o enables the channel.
- Collision: the inherited emulator also interprets /p as a comma-separated peripheral-ID list. A supplied port number is therefore also attempted as a peripheral ID during initialization.

Examples:

```
ZXSpectrumEmulator.exe /o  
ZXSpectrumEmulator.exe /o /p61000
```

Consequence: The remote server listens on the selected port. When /p is present, the same value is also attempted as a peripheral ID and normally produces a log warning.

/h

Meaning: Prints the executable help and prevents the emulation from starting.

General form:

```
/h
```

- No value.

Examples:

```
ZXSpectrumEmulator.exe /h
```

Consequence: Initialization returns failure after printing the help; the current main program therefore exits with code 1.

/f

Meaning: Loads a binary file whose header contains the destination address.

General form:

```
/fFILE
```

- FILE: path to the header-addressed binary file; attach it directly to /f.

Examples:

```
ZXSpectrumEmulator.exe /fprogram.bin  
ZXSpectrumEmulator.exe /fgames\demo.bin /a$080d
```

Consequence: The file is loaded before execution. An unreadable or invalid file aborts initialization.

/k

Meaning: Loads a block-structured memory file.

General form:

```
/kFILE
```

- FILE: path to a block file understood by the framework.

Examples:

```
ZXSpectrumEmulator.exe /ksnapshot.blocks
```

Consequence: Every encoded block is placed at its stored address. A load error aborts initialization.

/c

Meaning: Parses, assembles and loads an assembler source file.

General form:

```
/cFILE
```

- FILE: assembler source path.

Examples:

```
ZXSpectrumEmulator.exe /cstartup.asm /s
```


Consequence: The assembled bytes are installed before the main loop. Syntax or load errors are logged and initialization stops.

/l

Meaning: Selects the framework diagnostic level.

General form:

```
/lLEVEL
```

- 0: no diagnostics.
- 1: errors (default).
- 2: dump at exit.
- 3: errors and warnings.
- 4: all regular diagnostics.
- 5: internal trace diagnostics.

Examples:

```
ZXSpectrumEmulator.exe /l0  
ZXSpectrumEmulator.exe /l4
```

Consequence: Higher values produce progressively more diagnostic output and may reduce performance.

/a

Meaning: Overrides the address at which execution begins.

General form:

```
/aADDRESS
```

- ADDRESS: ZX Spectrum address in the framework number syntax, for example decimal or hexadecimal with \$.

Examples:

```
ZXSpectrumEmulator.exe /a$080d  
ZXSpectrumEmulator.exe /fprogram.bin /a2061
```

Consequence: A non-zero, valid address replaces the normal reset/program entry address.

/d

Meaning: Creates startup stop points at one or more addresses.

General form:

```
/dADDRESS1,ADDRESS2,...
```

- ADDRESSn: comma-separated address. Each entry accepts the framework number syntax.

Examples:

```
ZXSpectrumEmulator.exe /d$080d  
ZXSpectrumEmulator.exe /d$080d,$ffd2 /s
```

Consequence: Execution stops when the CPU reaches any listed address, allowing later inspection or continuation.

/s

Meaning: Controls whether the emulated computer starts stopped.

General form:

```
/s[YES|NO]
```

- YES or an empty value: start stopped.
- NO (and any non-YES non-empty value): start running.
- Omitted: start running.

Examples:

```
ZXSpectrumEmulator.exe /s  
ZXSpectrumEmulator.exe /sNO
```

Consequence: When enabled, the machine is initialized but the CPU does not run until a RUN/NEXT command or the equivalent UI action is issued.

/i

Meaning: Selects the ZX Spectrum ROM language.

General form:

```
/iLANG
```

- ENG: English (default).
- ESP: Spanish.
- NOR: Nordic.
- ARA: Arabic.
- Language identifiers are case-sensitive; use the uppercase forms above. Any other value loads the English ROM.

Examples:

```
ZXSpectrumEmulator.exe /iESP  
ZXSpectrumEmulator.exe /iARA
```

Consequence: The selected 16 KiB 48K-family ROM image is loaded before initialization; the screen's glyph reference remains sourced from the English ROM file.

/r

Meaning: Selects the initial ULA sound-output state.

General form:

```
/r[YES|NO]
```

- YES or an empty value: sound enabled.
- NO (and any other non-YES non-empty value): sound disabled.
- Omitted: sound enabled.

Examples:

```
ZXSpectrumEmulator.exe /r  
ZXSpectrumEmulator.exe /rNO
```

Consequence: The ZX Spectrum sound device attached to the ULA sound function starts enabled or disabled as requested.

/g

Meaning: Controls the CRT post-processing effect at startup.

General form:

```
/g[YES|NO]
```

- YES or an empty value: CRT effect enabled.
- NO (and any non-YES non-empty value): disabled.
- Omitted: disabled.

Examples:

```
ZXSpectrumEmulator.exe /g  
ZXSpectrumEmulator.exe /gNO
```

Consequence: The screen starts with or without CRT emulation. CRTON and CRTOFF can change it later.

/n

Meaning: Selects NTSC timing and screen geometry instead of PAL.

General form:

```
/n
```

- No value: the presence of the option selects NTSC; omission selects PAL.

Examples:

```
ZXSpectrumEmulator.exe /n
```

Consequence: The factory creates the NTSC ULA, NTSC screen and 3.5275 MHz clock. Without /n it creates the PAL ULA, PAL screen and 3.5 MHz clock.

/w

Meaning: Selects the installed RAM configuration.

General form:

```
/wCONF
```

- 0 or omitted: 16 KiB RAM configuration (default).
- 1: 48 KiB RAM configuration.
- 2 or any value greater than 1: unsupported. The current range check can pass or clamp it to the invalid internal value 2, which may assert in debug builds and must not be used.

Examples:

```
ZXSpectrumEmulator.exe /w0
```

```
ZXSpectrumEmulator.exe /w1
```

Consequence: Configuration 0 maps 16 KiB RAM with unmapped upper memory; configuration 1 maps the full 48 KiB RAM area before ROM startup.

/b

Meaning: Draws the diagnostic character-cell grid over the active screen and selects its startup colour.

General form:

```
/b[COLOR]
```

- COLOR omitted or 0: colour 0 (default).
- 1 through 3: the requested startup grid colour.
- Any value greater than 3: clamped to 3 by the startup parser.

Examples:

```
ZXSpectrumEmulator.exe /b  
ZXSpectrumEmulator.exe /b3
```

Consequence: The screen starts with its diagnostic 8x8 grid enabled. The later GRIDON console command has a wider 0-15 renderer range.

/m

Meaning: Selects the ZX Spectrum machine family implemented by this emulator.

General form:

```
/mMACHINE
```

- ZXSTD: standard 16K/48K ZX Spectrum; this is also the default.
- Any other value: silently falls back to ZXSTD in the current implementation.

Examples:

```
ZXSpectrumEmulator.exe /mZXSTD  
ZXSpectrumEmulator.exe /mZXSTD /w1 /iESP
```

Consequence: Only the original standard ZX Spectrum type is created. This selector does not provide 128K, +2, +2A or +3 models; /w, /i and /n independently select RAM, ROM language and video standard.

2. Startup Parameters: Version With a Local Console

Executable: ZX SpectrumEmulatorC.exe. Its main program delegates the complete command line to ZX SPECTRUM::ZX SpectrumEmulator, so all inherited framework options and ZX Spectrum-specific options are active even though the short main-level banner only advertises /h. It has no /o listener; commands are entered in its local console.

/h

Meaning: Prints the executable help and prevents the emulation from starting.

General form:

```
/h
```

- No value.

Examples:

```
ZXSpectrumEmulatorC.exe /h
```

Consequence: Initialization returns failure after printing the help; the current main program therefore exits with code 1.

/f

Meaning: Loads a binary file whose header contains the destination address.

General form:

```
/fFILE
```

- FILE: path to the header-addressed binary file; attach it directly to /f.

Examples:

```
ZXSpectrumEmulatorC.exe /fprogram.bin  
ZXSpectrumEmulatorC.exe /fgames\demo.bin /a$080d
```

Consequence: The file is loaded before execution. An unreadable or invalid file aborts initialization.

/k

Meaning: Loads a block-structured memory file.

General form:

```
/kFILE
```

- FILE: path to a block file understood by the framework.

Examples:

```
ZXSpectrumEmulatorC.exe /ksnapshot.blocks
```

Consequence: Every encoded block is placed at its stored address. A load error aborts initialization.

/c

Meaning: Parses, assembles and loads an assembler source file.

General form:

```
/cFILE
```

- FILE: assembler source path.

Examples:

```
ZXSpectrumEmulatorC.exe /cstartup.asm /s
```

Consequence: The assembled bytes are installed before the main loop. Syntax or load errors are logged and initialization stops.

/l

Meaning: Selects the framework diagnostic level.

General form:

```
/lLEVEL
```

- 0: no diagnostics.
- 1: errors (default).
- 2: dump at exit.
- 3: errors and warnings.
- 4: all regular diagnostics.
- 5: internal trace diagnostics.

Examples:

```
ZXSpectrumEmulatorC.exe /l0  
ZXSpectrumEmulatorC.exe /l4
```

Consequence: Higher values produce progressively more diagnostic output and may reduce performance.

/a

Meaning: Overrides the address at which execution begins.

General form:

```
/aADDRESS
```

- ADDRESS: ZX Spectrum address in the framework number syntax, for example decimal or hexadecimal with \$.

Examples:

```
ZXSpectrumEmulatorC.exe /a$080d  
ZXSpectrumEmulatorC.exe /fprogram.bin /a2061
```

Consequence: A non-zero, valid address replaces the normal reset/program entry address.

/d

Meaning: Creates startup stop points at one or more addresses.

General form:

```
/dADDRESS1,ADDRESS2,...
```

- ADDRESSn: comma-separated address. Each entry accepts the framework number syntax.

Examples:

```
ZXSpectrumEmulatorC.exe /d$080d  
ZXSpectrumEmulatorC.exe /d$080d,$ffd2 /s
```

Consequence: Execution stops when the CPU reaches any listed address, allowing later inspection or continuation.

/s

Meaning: Controls whether the emulated computer starts stopped.

General form:

```
/s[YES|NO]
```

- YES or an empty value: start stopped.
- NO (and any non-YES non-empty value): start running.
- Omitted: start running.

Examples:

```
ZXSpectrumEmulatorC.exe /s  
ZXSpectrumEmulatorC.exe /sNO
```

Consequence: When enabled, the machine is initialized but the CPU does not run until a RUN/NEXT command or the equivalent UI action is issued.

/i

Meaning: Selects the ZX Spectrum ROM language.

General form:

```
/iLANG
```

- ENG: English (default).
- ESP: Spanish.
- NOR: Nordic.
- ARA: Arabic.
- Language identifiers are case-sensitive; use the uppercase forms above. Any other value loads the English ROM.

Examples:

```
ZXSpectrumEmulatorC.exe /iESP  
ZXSpectrumEmulatorC.exe /iARA
```

Consequence: The selected 16 KiB 48K-family ROM image is loaded before initialization; the screen's glyph reference remains sourced from the English ROM file.

/r

Meaning: Selects the initial ULA sound-output state.

General form:

```
/r[YES|NO]
```

- YES or an empty value: sound enabled.
- NO (and any other non-YES non-empty value): sound disabled.
- Omitted: sound enabled.

Examples:

```
ZXSpectrumEmulatorC.exe /r  
ZXSpectrumEmulatorC.exe /rNO
```

Consequence: The ZX Spectrum sound device attached to the ULA sound function starts enabled or disabled as requested.

/p

Meaning: Connects one or more peripherals during startup.

General form:

```
/pPER1[,PER2...]
```

- 10: typewriter keyboard input.
- 100: Sinclair cassette signal simulation.
- 101: ZX Spectrum TAP/TZX cassette-data injection.
- 102: ZX Printer thermal-printer simulation.
- 201: cartridge implementation supported by the builder, although it is not listed by POSSIBLEPERS.

Examples:

```
ZXSpectrumEmulatorC.exe /p10  
ZXSpectrumEmulatorC.exe /p100,102
```

Consequence: Each requested peripheral is built and connected to the first compatible device during initialization. Unsupported IDs are logged and initialization continues.

/g

Meaning: Controls the CRT post-processing effect at startup.

General form:

```
/g[YES|NO]
```

- YES or an empty value: CRT effect enabled.
- NO (and any non-YES non-empty value): disabled.
- Omitted: disabled.

Examples:


```
ZXSpectrumEmulatorC.exe /g
ZXSpectrumEmulatorC.exe /gNO
```

Consequence: The screen starts with or without CRT emulation. CRTON and CRTOFF can change it later.

/n

Meaning: Selects NTSC timing and screen geometry instead of PAL.

General form:

```
/n
```

- No value: the presence of the option selects NTSC; omission selects PAL.

Examples:

```
ZXSpectrumEmulatorC.exe /n
```

Consequence: The factory creates the NTSC ULA, NTSC screen and 3.5275 MHz clock. Without /n it creates the PAL ULA, PAL screen and 3.5 MHz clock.

/w

Meaning: Selects the installed RAM configuration.

General form:

```
/wCONF
```

- 0 or omitted: 16 KiB RAM configuration (default).
- 1: 48 KiB RAM configuration.
- 2 or any value greater than 1: unsupported. The current range check can pass or clamp it to the invalid internal value 2, which may assert in debug builds and must not be used.

Examples:

```
ZXSpectrumEmulatorC.exe /w0
ZXSpectrumEmulatorC.exe /w1
```

Consequence: Configuration 0 maps 16 KiB RAM with unmapped upper memory; configuration 1 maps the full 48 KiB RAM area before ROM startup.

/b

Meaning: Draws the diagnostic character-cell grid over the active screen and selects its startup colour.

General form:

```
/b[COLOR]
```

- COLOR omitted or 0: colour 0 (default).
- 1 through 3: the requested startup grid colour.
- Any value greater than 3: clamped to 3 by the startup parser.

Examples:

```
ZXSpectrumEmulatorC.exe /b  
ZXSpectrumEmulatorC.exe /b3
```

Consequence: The screen starts with its diagnostic 8x8 grid enabled. The later GRIDON console command has a wider 0-15 render range.

/m

Meaning: Selects the ZX Spectrum machine family implemented by this emulator.

General form:

```
/mMACHINE
```

- ZXSTD: standard 16K/48K ZX Spectrum; this is also the default.
- Any other value: silently falls back to ZXSTD in the current implementation.

Examples:

```
ZXSpectrumEmulatorC.exe /mZXSTD  
ZXSpectrumEmulatorC.exe /mZXSTD /w1 /iESP
```

Consequence: Only the original standard ZX Spectrum type is created. This selector does not provide 128K, +2, +2A or +3 models; /w, /i and /n independently select RAM, ROM language and video standard.

3. Command Console Reference

The local console checks its eleven emulator-level commands first. All remaining text is delegated through ZXSPECTRUM::CommandBuilder, then SINCLAIR::CommandBuilder, and finally MCHEmul::StandardCommandBuilder. The remote /o channel in ZXspectrumEmulator.exe uses only that three-builder chain. The availability line in each entry makes this boundary explicit.

3.1 Standard Framework Commands

HELP

Availability: Local console and remote /o channel.

Accepted aliases: CHELP.

Meaning: Displays the ZX Spectrum console help assembled from the ZX Spectrum, Sinclair, local-console and standard help files.

General syntax:

```
HELP
```

Parameters:

- No parameters.

Examples:

```
HELP
```

Result and consequences: Displays the ZX Spectrum console help assembled from the ZX Spectrum, Sinclair, local-console and standard help files.

AUTHOR

Availability: Local console and remote /o channel.

Accepted aliases: CAUTHOR.

Meaning: Returns framework author information.

General syntax:

```
AUTHOR
```

Parameters:

- No parameters.

Examples:

```
AUTHOR
```

Result and consequences: Returns framework author information.

STATUS

Availability: Local console and remote /o channel.

Accepted aliases: CSTATUS.

Meaning: Shows the Z80 status register.

General syntax:

```
STATUS
```

Parameters:

- No parameters.

Examples:

```
STATUS
```

Result and consequences: Shows the Z80 status register.

PC

Availability: Local console and remote /o channel.

Accepted aliases: CPC.

Meaning: Shows the current program counter.

General syntax:

```
PC
```

Parameters:

- No parameters.

Examples:

```
PC
```

Result and consequences: Shows the current program counter.

REGISTERS

Availability: Local console and remote /o channel.

Accepted aliases: CREGISTERS.

Meaning: Shows all CPU registers.

General syntax:

```
REGISTERS
```

Parameters:

- No parameters.

Examples:

```
REGISTERS
```

Result and consequences: Shows all CPU registers.

CHANGereg

Availability: Local console and remote /o channel.

Accepted aliases: CCHANGereg.

Meaning: Changes one CPU internal register.

General syntax:

```
CHANGereg REGNAME VALUE
```

Parameters:

- REGNAME: registered Z80 register name.
- VALUE: new decimal, z-prefixed binary, o-prefixed octal or \$-prefixed hexadecimal value.

Examples:

```
CHANGereg A $41
```

Result and consequences: The register changes only if it exists and accepts the supplied byte width; otherwise an error is returned.

STACK

Availability: Local console and remote /o channel.

Accepted aliases: CSTACK.

Meaning: Shows stack state.

General syntax:

```
STACK [ALL]
```

Parameters:

- ALL: optional; includes stack contents in addition to basic state.

Examples:

```
STACK  
STACK ALL
```

Result and consequences: Shows stack state.

CPUSTATUS

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSTATUS.

Meaning: Shows registers, status, program counter and stack.

General syntax:

```
CPUSTATUS
```

Parameters:

- No parameters.

Examples:

```
CPUSTATUS
```

Result and consequences: Shows registers, status, program counter and stack.

CPUSTATUS

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSSTATUS.

Meaning: Shows the compact CPU status without stack detail.

General syntax:

```
CPUSSTATUS
```

Parameters:

- No parameters.

Examples:

```
CPUSSTATUS
```

Result and consequences: Shows the compact CPU status without stack detail.

CPUSTATE

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSTATE.

Meaning: Shows the CPU execution state and state-specific data.

General syntax:

```
CPUSTATE
```

Parameters:

- No parameters.

Examples:

```
CPUSTATE
```

Result and consequences: Shows the CPU execution state and state-specific data.

CPUINFO

Availability: Local console and remote /o channel.

Accepted aliases: CCPUINFO.

Meaning: Returns the complete CPU information structure.

General syntax:

```
CPUINFO
```

Parameters:

- No parameters.

Examples:

```
CPUINFO
```

Result and consequences: Returns the complete CPU information structure.

MEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORY.

Meaning: Reads the active CPU-visible memory range.

General syntax:

```
MEMORY INITIAL [FINAL]
```

Parameters:

- INITIAL: first address.
- FINAL: optional last address; with no final address only the initial location is reported.

Examples:

```
MEMORY $0400  
MEMORY $0400 $0427
```

Result and consequences: Reads the active CPU-visible memory range.

MEMORYDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYDUMP.

Meaning: Dumps every memory view between two addresses, including inactive RAM/ROM views.

General syntax:

```
MEMORYDUMP INITIAL FINAL
```

Parameters:

- INITIAL: first address.
- FINAL: last address.

Examples:

```
MEMORYDUMP $a000 $a03f
```

Result and consequences: Dumps every memory view between two addresses, including inactive RAM/ROM views.

MEMORYSTR

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYSTR.

Meaning: Shows the configured memory structure.

General syntax:

```
MEMORYSTR [ALL]
```

Parameters:

- ALL: optional; includes inactive regions. Without it only active regions are shown.

Examples:

```
MEMORYSTR  
MEMORYSTR ALL
```

Result and consequences: Shows the configured memory structure.

SETMEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CSETMEMORY.

Meaning: Writes a value to one address or repeatedly across a range.

General syntax:

```
SETMEMORY INITIAL [FINAL] VALUE
```

Parameters:

- INITIAL: first address.
- FINAL: optional last address.
- VALUE: byte sequence in framework numeric syntax; multi-byte values advance by their own length.

Examples:

```
SETMEMORY $0400 $41  
SETMEMORY $0400 $0427 $20
```

Result and consequences: The active memory view is modified immediately; unsafe writes can corrupt state or crash the emulated program.

MEMORYANALYSIS

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYANALYSIS.

Meaning: Analyzes the current memory configuration.

General syntax:

```
MEMORYANALYSIS
```

Parameters:

- No parameters.

Examples:

```
MEMORYANALYSIS
```

Result and consequences: Analyzes the current memory configuration.

STOP

Availability: Local console and remote /o channel.

Accepted aliases: CSTOP.

Meaning: Stops CPU execution.

General syntax:

```
STOP
```

Parameters:

- No parameters.

Examples:

```
STOP
```

Result and consequences: Stops CPU execution.

RUN

Availability: Local console and remote /o channel.

Accepted aliases: R, CRUN.

Meaning: Starts or resumes CPU execution; alias R.

General syntax:

```
RUN [ADDRESS]
```

Parameters:

- ADDRESS: optional new program counter before running.

Examples:

```
RUN  
R $080d
```

Result and consequences: The CPU resumes immediately, optionally after changing PC.

SETPC

Availability: Local console and remote /o channel.

Accepted aliases: CSETPC.

Meaning: Sets the program counter without starting execution.

General syntax:

```
SETPC ADDRESS
```

Parameters:

- ADDRESS: new CPU address.

Examples:

```
SETPC $080d
```

Result and consequences: Subsequent execution continues from the new address; an unsafe address can break the program.

NEXT

Availability: Local console and remote /o channel.

Accepted aliases: N, CNEXT.

Meaning: Executes the next instruction; alias N.

General syntax:

```
NEXT
```

Parameters:

- No parameters.

Examples:

```
NEXT  
N
```

Result and consequences: Executes the next instruction; alias N.

NEXTSTACK

Availability: Local console and remote /o channel.

Accepted aliases: NS, CNEXTSTACK.

Meaning: Runs until the stack pointer returns to its initial value; alias NS.

General syntax:

```
NEXTSTACK
```

Parameters:

- No parameters.

Examples:

```
NS
```

Result and consequences: Useful for stepping over a routine, although behavior depends on the routine's stack discipline.

SHOWNEXT

Availability: Local console and remote /o channel.

Accepted aliases: SN, CSHOWNEXT.

Meaning: Disassembles upcoming instruction(s); alias SN.

General syntax:

```
SHOWNEXT [NUMBER]
```

Parameters:

- NUMBER: optional count; one instruction by default.

Examples:

```
SHOWNEXT  
SN 8
```

Result and consequences: Disassembles upcoming instruction(s); alias SN.

INST

Availability: Local console and remote /o channel.

Accepted aliases: CINST.

Meaning: Shows the last instruction executed.

General syntax:

```
INST
```

Parameters:

- No parameters.

Examples:

```
INST
```

Result and consequences: Shows the last instruction executed.

BREAKS

Availability: Local console and remote /o channel.

Accepted aliases: CBREAKS.

Meaning: Lists all stop-point addresses.

General syntax:

```
BREAKS
```

Parameters:

- No parameters.

Examples:

```
BREAKS
```

Result and consequences: Lists all stop-point addresses.

SETBREAK

Availability: Local console and remote /o channel.

Accepted aliases: CSETBREAK.

Meaning: Adds one or more breakpoints.

General syntax:

```
SETBREAK ADDRESS1 [ADDRESS2 ...]
```

Parameters:

- ADDRESSn: one or more addresses in framework numeric syntax.

Examples:

```
SETBREAK $080d $ffd2
```

Result and consequences: Adds one or more breakpoints.

REMOVEBREAK

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEBREAK.

Meaning: Removes selected breakpoints.

General syntax:

```
REMOVEBREAK ADDRESS1 [ADDRESS2 ...]
```

Parameters:

- ADDRESSn: one or more existing breakpoint addresses.

Examples:

```
REMOVEBREAK $ffd2
```

Result and consequences: Removes selected breakpoints.

REMOVEBREAKS

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEBREAKS.

Meaning: Removes every breakpoint.

General syntax:

```
REMOVEBREAKS
```

Parameters:

- No parameters.

Examples:

```
REMOVEBREAKS
```

Result and consequences: Removes every breakpoint.

SPEED

Availability: Local console and remote /o channel.

Accepted aliases: CSPEED.

Meaning: Shows measured CPU and screen-I/O execution speed.

General syntax:

```
SPEED
```

Parameters:

- No parameters.

Examples:

```
SPEED
```

Result and consequences: Shows measured CPU and screen-I/O execution speed.

LOADBIN

Availability: Local console and remote /o channel.

Accepted aliases: CLOADBIN.

Meaning: Loads a raw binary file at an explicit address.

General syntax:

```
LOADBIN FILE ADDRESS
```

Parameters:

- FILE: raw binary path.
- ADDRESS: first destination address.

Examples:

```
LOADBIN charset.bin $2000
```

Result and consequences: Bytes are copied into memory without an embedded load-address header.

SAVEBIN

Availability: Local console and remote /o channel.

Accepted aliases: CSAVEBIN.

Meaning: Saves a memory range as raw binary.

General syntax:

```
SAVEBIN FILE SIZE ADDRESS
```

Parameters:

- FILE: output path.
- SIZE: byte count.
- ADDRESS: first source address.

Examples:

```
SAVEBIN screen.bin 1000 $0400
```

Result and consequences: Saves a memory range as raw binary.

DEBUGON

Availability: Local console and remote /o channel.

Accepted aliases: CACTIVATEDEEPDEBUG.

Meaning: Enables deep execution debugging.

General syntax:

```
DEBUGON FILE CPU [APPEND] [FILTERS...]
```

Parameters:

- FILE: output log.
- CPU: YES includes CPU execution; NO omits it.
- APPEND: optional YES appends, NO overwrites (default).
- D:START-END or ALLDIR: address filter.
- C:IDs or ALLCHIP: chip filter.
- I:IDs or ALLIO: device filter.
- M:IDs or ALLMEM: memory filter.

Examples:

```
DEBUGON zxspectrum.log YES NO D:$0800-$0fff ALLCHIP ALLIO ALLMEM
```

Result and consequences: Tracing begins with the selected sources and can substantially slow emulation. It cannot coexist with interrupt-only debugging.

DEBUGOFF

Availability: Local console and remote /o channel.

Accepted aliases: CDESACTIVATEDEEPDEBUG.

Meaning: Stops deep execution debugging.

General syntax:

```
DEBUGOFF
```

Parameters:

- No parameters.

Examples:

```
DEBUGOFF
```

Result and consequences: Stops deep execution debugging.

IDEBUGON

Availability: Local console and remote /o channel.

Accepted aliases: CIDEBUGON.

Meaning: Enables interrupt-only debugging.

General syntax:

```
IDEBUGON FILE [APPEND]
```

Parameters:

- FILE: output log.
- APPEND: optional YES appends; NO overwrites (default).

Examples:

```
IDEBUGON interrupts.log NO
```

Result and consequences: Interrupt tracing begins and cannot coexist with deep debugging.

IDEBUGOFF

Availability: Local console and remote /o channel.

Accepted aliases: CIDEBUGOFF.

Meaning: Stops interrupt-only debugging.

General syntax:

```
IDEBUGOFF
```

Parameters:

- No parameters.

Examples:

```
IDEBUGOFF
```

Result and consequences: Stops interrupt-only debugging.

RESTART

Availability: Local console and remote /o channel.

Accepted aliases: CRESTART.

Meaning: Requests a restart and exits the current computer loop so it can be rebuilt.

General syntax:

```
RESTART STOPPED [LEVEL]
```

Parameters:

- STOPPED: YES restarts stopped; NO restarts running.
- LEVEL: optional 0 CPU-only/mild restart (default), 1 computer restart preserving memory, greater than 1 full restart.

Examples:

```
RESTART YES 0
RESTART NO 2
```

Result and consequences: Requests a restart and exits the current computer loop so it can be rebuilt.

DEVICES

Availability: Local console and remote /o channel.

Accepted aliases: CDEVICES.

Meaning: Shows all or selected I/O devices.

General syntax:

```
DEVICES [ID1 ID2 ...]
```

Parameters:

- IDn: optional device identifiers; with none, all devices are returned.

Examples:

```
DEVICES
DEVICES 1 2
```

Result and consequences: Shows all or selected I/O devices.

PERIPHERALS

Availability: Local console and remote /o channel.

Accepted aliases: CPERIPHERALS.

Meaning: Shows all or selected connected peripherals.

General syntax:

```
PERIPHERALS [ID1 ID2 ...]
```

Parameters:

- IDn: optional peripheral identifiers; with none, all are returned.

Examples:

```
PERIPHERALS
PERIPHERALS 100
```

Result and consequences: Shows all or selected connected peripherals.

PERCOMMAND

Availability: Local console and remote /o channel.

Accepted aliases: CPERCMD.

Meaning: Sends a device-specific command to a connected peripheral.

General syntax:

```
PERCOMMAND PERID CMDID [ATTRS...]
```

Parameters:

- PERID: connected peripheral ID.
- CMDID: command ID implemented by that peripheral.
- ATTRS: optional positional attributes forwarded unchanged.

Examples:

```
PERCOMMAND 100 0
```

Result and consequences: The peripheral executes the request if both IDs and its parameter contract are valid; otherwise the answer reports an error.

ASSIGNJOY

Availability: Local console and remote /o channel.

Accepted aliases: CASSIGNJ.

Meaning: Remaps a joystick identifier.

General syntax:

```
ASSIGNJOY ID NEWID
```

Parameters:

- ID: current joystick ID.
- NEWID: required target ID; using the same ID removes an existing assignment.

Examples:

```
ASSIGNJOY 0 1
```

Result and consequences: Remaps a joystick identifier.

CLOCKFACTOR

Availability: Local console and remote /o channel.

Accepted aliases: CCLOCKFACTOR.

Meaning: Changes the CPU clock multiplier.

General syntax:

```
CLOCKFACTOR VALUE
```

Parameters:

- VALUE: numeric factor; 1 means normal speed.

Examples:

```
CLOCKFACTOR 2
```

Result and consequences: The emulation pacing target changes; values above one request faster execution.

SOUNDON

Availability: Local console and remote /o channel.

Accepted aliases: CSOUNDON.

Meaning: Enables ZX Spectrum ULA sound output.

General syntax:

```
SOUNDON
```

Parameters:

- No parameters.

Examples:

```
SOUNDON
```

Result and consequences: Enables the sound system attached to the ULA sound function.

SOUNDOFF

Availability: Local console and remote /o channel.

Accepted aliases: CSOUNDOFF.

Meaning: Disables ZX Spectrum ULA sound output.

General syntax:

```
SOUNDOFF
```

Parameters:

- No parameters.

Examples:

```
SOUNDOFF
```

Result and consequences: Disables the sound system; emulation continues without audible ULA output.

INTERRUPTS

Availability: Local console and remote /o channel.

Accepted aliases: CINTERRUPTS.

Meaning: Shows all interrupts known by the computer.

General syntax:

```
INTERRUPTS
```

Parameters:

- No parameters.

Examples:

```
INTERRUPTS
```

Result and consequences: Shows all interrupts known by the computer.

INTSET

Availability: Local console and remote /o channel.

Accepted aliases: CINTSET.

Meaning: Enables or disables one interrupt or every interrupt.

General syntax:

```
INTSET ID|ALL ON|OFF
```

Parameters:

- ID or ALL: target interrupt.
- ON or OFF: requested state.

Examples:

```
INTSET ALL ON
```

Result and consequences: An unknown interrupt ID produces an error.

CHIPS

Availability: Local console and remote /o channel.

Accepted aliases: CCHIPS.

Meaning: Lists the emulated chips and their identifying attributes.

General syntax:

```
CHIPS
```

Parameters:

- No parameters.

Examples:

```
CHIPS
```

Result and consequences: Lists the emulated chips and their identifying attributes.

CRTON

Availability: Local console and remote /o channel.

Accepted aliases: CCRTON.

Meaning: Enables the CRT screen effect.

General syntax:

```
CRTON
```

Parameters:

- No parameters.

Examples:

```
CRTON
```

Result and consequences: Enables the CRT screen effect.

CRTOFF

Availability: Local console and remote /o channel.

Accepted aliases: CCRTOFF.

Meaning: Disables the CRT screen effect.

General syntax:

```
CRTOFF
```

Parameters:

- No parameters.

Examples:

```
CRTOFF
```

Result and consequences: Disables the CRT screen effect.

DATASETTE

Availability: Local console and remote /o channel.

Accepted aliases: CDATASETTE.

Meaning: Shows datasette state when a supported datasette is connected.

General syntax:

```
DATASETTE
```

Parameters:

- No parameters.

Examples:

```
DATASETTE
```

Result and consequences: Shows datasette state when a supported datasette is connected.

GRIDON

Availability: Local console and remote /o channel.

Accepted aliases: CGRIDON.

Meaning: Enables the diagnostic grid overlay on the screen.

General syntax:

```
GRIDON COLOR
```

Parameters:

- COLOR: mandatory numeric grid color.

Examples:

```
GRIDON 14
```

Result and consequences: Enables the standard screen grid with the requested color.

GRIDOFF

Availability: Local console and remote /o channel.

Accepted aliases: CGRIDOFF.

Meaning: Disables the diagnostic grid overlay.

General syntax:

```
GRIDOFF
```

Parameters:

- No parameters.

Examples:

```
GRIDOFF
```

Result and consequences: Disables the standard screen grid.

PICTURE

Availability: Local console and remote /o channel.

Accepted aliases: CPICTURE.

Meaning: Saves a screenshot of the emulator screen.

General syntax:

```
PICTURE FILE
```

Parameters:

- FILE: output PNG, JPG or BMP filename.

Examples:

```
PICTURE zxspectrum-screen.png
```

Result and consequences: A supported extension writes the current screen; unsupported formats fail.

SETHOOK

Availability: Local console and remote /o channel.

Accepted aliases: CSETHOOK.

Meaning: Creates an execution hook.

General syntax:

```
SETHOOK ID TYPE [PARAMS...]
```

Parameters:

- ID: unique hook identifier.
- TYPE: hook kind; currently 0 stops on memory read/write.
- For TYPE 0, PARAMS are mandatory INITIAL and optional FINAL addresses.

Examples:

```
SETHOOK 1 0 $d000 $d3ff
```

Result and consequences: Creates an execution hook.

REMOVEHOOK

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEHOOK.

Meaning: Removes a hook by identifier.

General syntax:

```
REMOVEHOOK ID
```

Parameters:

- ID: hook identifier.

Examples:

```
REMOVEHOOK 1
```

Result and consequences: If the hook does not exist, no state changes.

HOOKS

Availability: Local console and remote /o channel.

Accepted aliases: CHOOKS.

Meaning: Lists active hooks.

General syntax:

```
HOOKS
```

Parameters:

- No parameters.

Examples:

```
HOOKS
```

Result and consequences: Lists active hooks.

HOOKSHELP

Availability: Local console and remote /o channel.

Accepted aliases: CHOOKSHELP.

Meaning: Shows help for supported hook types.

General syntax:

```
HOOKSHELP
```

Parameters:

- No parameters.

Examples:

```
HOOKSHELP
```

Result and consequences: Shows help for supported hook types.

CMOVEPARAMS

Availability: Local console and remote /o channel.

Accepted aliases: No public alias.

Meaning: Builds the framework's internal MoveParametersToAnswerCommand.

General syntax:

```
CMOVEPARAMS
```

Parameters:

- No parameters are executable under the current command contract.

Examples:

```
CMOVEPARAMS
```

Result and consequences: The command returns an empty answer when invoked directly; it is an internal response helper and is not advertised in the help file.

3.2 Inherited Sinclair Commands

SYSVARS

Availability: Local console and remote /o channel.

Accepted aliases: CSYSVARS.

Meaning: Returns every system variable defined for the selected ZX Spectrum ROM profile.

General syntax:

```
SYSVARS
```

Parameters:

- No parameters.

Examples:

```
SYSVARS
```

Result and consequences: Returns the current values using the system-variable definitions loaded for the selected ROM language.

SYSVAR

Availability: Local console and remote /o channel.

Accepted aliases: CSYSVAR.

Meaning: Returns one named ZX Spectrum system variable.

General syntax:

```
SYSVAR VNAME
```

Parameters:

- VNAME: mandatory variable name from the active ZX Spectrum system-variable table.

Examples:

```
SYSVAR FRAMES  
SYSVAR RAMTOP
```

Result and consequences: Returns the named variable and its current memory value; an unknown name returns an error entry.

3.3 ZX Spectrum-Specific Commands

ULA

Availability: Local console and remote /o channel.

Accepted aliases: CULA.

Meaning: Returns the current ZX Spectrum ULA and video-generation state.

General syntax:

```
ULA
```

Parameters:

- No parameters.

Examples:

```
ULA
```

Result and consequences: Returns the active ULA state, including raster and display-generation information for PAL or NTSC operation.

ULAEVENTS

Availability: Local console and remote /o channel.

Accepted aliases: CULAEVENTS.

Meaning: Enables or disables visualization of significant CPU/ULA events.

General syntax:

```
ULAEVENTS ON|OFF
```

Parameters:

- ON: enables event visualization.
- OFF: disables it. The implementation accepts any single token; only the exact value ON enables the feature.

Examples:

```
ULAEVENTS ON  
ULAEVENTS OFF
```

Result and consequences: Changes the event-overlay flag used by the active ZX Spectrum ULA.

SCREENDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CSCREENDUMP.

Meaning: Returns a snapshot of ZX Spectrum bitmap memory.

General syntax:

```
SCREENDUMP
```

Parameters:

- No parameters.

Examples:

```
SCREENDUMP
```

Result and consequences: Returns the current 6,144-byte bitmap area selected by the machine's screen-memory snapshot routine.

COLORDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CCOLORDUMP.

Meaning: Returns a snapshot of ZX Spectrum attribute memory.

General syntax:

```
COLORDUMP
```

Parameters:

- No parameters.

Examples:

```
COLORDUMP
```

Result and consequences: Returns the current 768-byte colour-attribute area selected by the machine's colour-memory snapshot routine.

CHARSDRAW

Availability: Local console and remote /o channel.

Accepted aliases: CCHARSDRAW.

Meaning: Returns textual drawings of all or selected ZX Spectrum character glyphs.

General syntax:

```
CHARSDRAW [0..95 ...]
```

Parameters:

- Character indexes: optional values from 0 through 95; out-of-range values are ignored and duplicates are coalesced. With none, all 96 glyphs are rendered.

Examples:

```
CHARSDRAW  
CHARSDRAW 0 31 64 95
```

Result and consequences: Returns textual 8x8 glyph drawings from the standard English 48K ROM character data used by the screen object.

3.4 Local-Console-Only Commands

LOADPRG

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads, assembles and installs an assembler source through the emulator object.

General syntax:

```
LOADPRG FILE
```

Parameters:

- FILE: assembler source path.

Examples:

```
LOADPRG demo.asm
```

Result and consequences: Compilation or loading errors are reported by the local console.

LOADBINARY

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads a binary containing its destination address in the file header.

General syntax:

```
LOADBINARY FILE
```

Parameters:

- FILE: header-addressed binary path.

Examples:

```
LOADBINARY demo.bin
```

Result and consequences: Loads a binary containing its destination address in the file header.

LOADBLOCKS

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads a block-structured memory file.

General syntax:

```
LOADBLOCKS FILE
```

Parameters:

- FILE: block file path.

Examples:

```
LOADBLOCKS state.blocks
```

Result and consequences: Loads a block-structured memory file.

DECOMPILE

Availability: Local console only.

Accepted aliases: None.

Meaning: Disassembles a memory region.

General syntax:

```
DECOMPILE ADDRESS BYTES
```

Parameters:

- ADDRESS: first address.
- BYTES: approximate byte count; output rounds to complete instructions.

Examples:

```
DECOMPILE $080d 64
```

Result and consequences: Disassembles a memory region.

POSSIBLEPERS

Availability: Local console only.

Accepted aliases: None.

Meaning: Lists the peripheral IDs and construction attributes advertised by the ZX Spectrum I/O peripheral builder.

General syntax:

```
POSSIBLEPERS
```

Parameters:

- No parameters.

Examples:

```
POSSIBLEPERS
```

Result and consequences: Lists IDs 10 (typewriter), 100 (cassette), 101 (cassette injection) and 102 (thermal printer). Cartridge ID 201 is accepted by the builder but is not advertised.

CONNECTPER

Availability: Local console only.

Accepted aliases: None.

Meaning: Builds and connects a peripheral to the first compatible ZX Spectrum device.

General syntax:

```
CONNECTPER ID [ATTRS...]
```

Parameters:

- ID: peripheral-emulation ID (10, 100, 101, 102 or the additionally supported cartridge ID 201).
- ATTRS: optional construction attributes. For printer 102, use f:FILENAME and p:THERMAL or p:THERMAL-PS.

Examples:

```
CONNECTPER 100  
CONNECTPER 102 f:printer.ps p:THERMAL-PS
```

Result and consequences: Connection succeeds only when the ID, attributes and a compatible target device are available; otherwise the answer reports an error.

DISCONNECTPERS

Availability: Local console only.

Accepted aliases: None.

Meaning: Disconnects one or more peripherals.

General syntax:

```
DISCONNECTPERS ID1 [ID2 ...]
```

Parameters:

- IDn: connected peripheral IDs.

Examples:

```
DISCONNECTPERS 100 101
```

Result and consequences: Disconnects one or more peripherals.

LOADPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads media or program data into a connected ZX Spectrum peripheral.

General syntax:

```
LOADPERDATA ID FILE [DEBUGFILE]
```

Parameters:

- ID: connected peripheral ID.
- FILE: supported data file; for example sampled TZX data for ID 100, or TAP/TAP-like TZX data for injection ID 101.
- DEBUGFILE: optional file that starts full-range format-related debugging after a successful load.

Examples:

```
LOADPERDATA 100 tape.tzx  
LOADPERDATA 101 game.tap load-trace.log
```

Result and consequences: The operation fails if the peripheral rejects the file type. Injection accepts TAP and only TZX files whose blocks are all TAP-compatible.

EMPTYPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Creates and attaches empty media data when the peripheral implements that operation.

General syntax:

```
EMPTYPERDATA ID FILE
```

Parameters:

- ID: connected peripheral ID.
- FILE: destination whose extension selects a writable format.

Examples:

```
EMPTYPERDATA 100 blank.tzx
```

Result and consequences: The Sinclair cassette can create an empty TZX container; unsupported peripherals or formats return an error.

SAVEPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Saves a peripheral's current media data.

General syntax:

```
SAVEPERDATA ID [FILE]
```

Parameters:

- ID: connected peripheral ID.
- FILE: optional output path; it may be omitted when a prior load/create operation established the filename.

Examples:

```
SAVEPERDATA 100  
SAVEPERDATA 100 copy.tzx
```

Result and consequences: Saves current cassette data when the peripheral can retrieve it and the selected format is writable. Injection ID 101 cannot retrieve data.

CLEARPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Clears all data currently loaded into peripherals.

General syntax:

```
CLEARPERDATA
```

Parameters:

- No parameters.

Examples:

```
CLEARPERDATA
```

Result and consequences: Commonly used to force a subsequent program/media reload.

Appendices

Appendix A. Command Availability Summary

The table distinguishes the interactive console from the command channel of the non-console executable. Remote availability assumes ZX Spectrum Emulator.exe was started with /o and that the client is connected.

Command family	ZXSpectrumEmulatorC local console	ZXSpectrumEmulator remote /o
Standard builder	Yes	Yes
Inherited Sinclair builder	Yes	Yes
ZX Spectrum builder	Yes	Yes
LocalConsole emulator/media operations	Yes	No

Appendix B. Source-of-Truth Maintenance Rule

This file is a maintained repository artifact. Every change that adds, removes, renames or changes a ZX Spectrum startup option, command-builder registration, LocalConsole-only command, syntax, parameter meaning, consequence or visible formatted output must include a review and, when applicable, an update of docs/ZXSpectrumData/ZXSpectrumEmulator_UserGuide.docx. Update the corresponding canonical .fmt source whenever the command's InfoStructure contract changes.

Audit the complete chain, not only the edited class: both ZXSpectrumEmulator entry points and the ZX SPECTRUM::ZXSpectrumEmulator hierarchy for startup options; LocalConsole for local-only commands; and ZX SPECTRUM::CommandBuilder -> SINCLAIR::CommandBuilder -> MCHEmul::StandardCommandBuilder for builder commands.

Illustrations

NO ILLUSTRATIONS ARE INCLUDED IN THIS EDITION.

IGNACIO CEA FORNIÉS

Equations

NO EQUATIONS ARE INCLUDED IN THIS EDITION.

IGNACIO CEA FORNIÉS